Audition MCF P13/LIPN

Ivan Yakovlev

2024 – 2026	Postdoc (MPI for Mathematics, Bonn)
2021 – 2024	Doctorat (LaBRI, Bordeaux)
2018 – 2021	ENS + M2 Math Fonda (Paris)
2015 – 2019	Licence Math Appli (Kiev)

Enseignement : expérience

2021 – 2024, Université de Bordeaux, Licence Info.

L1	Initiation à la programmation C	32h	CI/TDM
L1	Algorithmique des tableaux (Python)	32h	TDM
L2	Algorithmique des structures de données arborescentes (OCaml)	32h	CI/TDM
L2	Probabilités, Statistiques et Combinatoire	28h	TD
L2	Réseaux	32h	TD/TDM

Enseignement : projet d'intégration

Dès la rentrée :

- programmation impérative/fonctionnelle/orientée objet, réseaux, algorithmique, automates/langages, proba/stats;
- encadrement;

Á court terme :

- architecture, administration, systèmes d'exploitation, programmation web, bases de données;
- responsabilité d'année.

Activités diverses

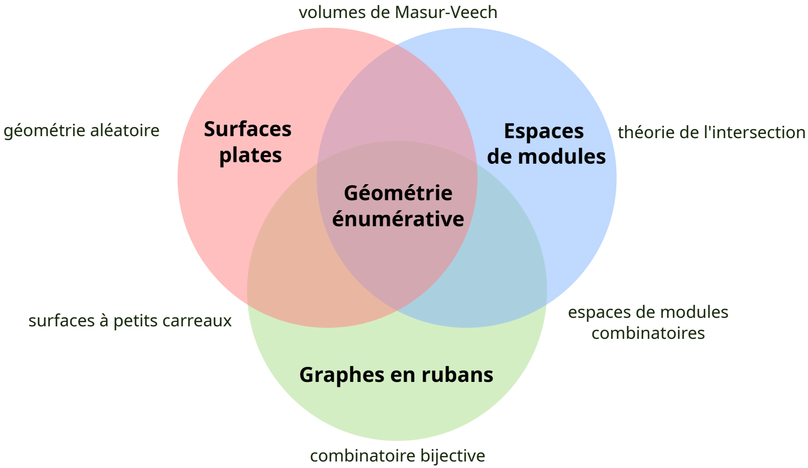
Organisation

- séminaire de l'équipe "Combinatoire et Interactions" (Bordeaux, 2022 – 2023)
- école "Surfaces plates et interactions" (Le Teich, 2024)
- groupe de lecture "Géométrie spectrale" (Bonn, 2026)

Diffusion

- enseignement des mathématiques olympiques aux collégiens et lycéens (Kiev 2015 – 2018, Marseille 2022)
- organisation / jury aux olympiades mathématiques (Kiev 2015 – 2018, EGMO 2019)

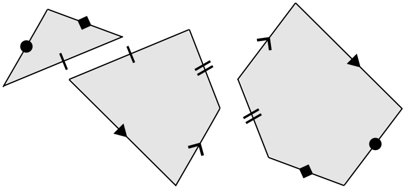
Recherche

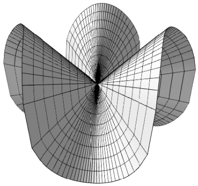

volumes de Masur-Veech
géométrie aléatoire
Surfaces plates
Espaces de modules
théorie de l'intersection
Géométrie énumérative
surfaces à petits carreaux
espaces de modules combinatoires
Graphes en rubans
combinatoire bijective

Travaux

1. I. Yakovlev, *Contribution of n-cylinder square-tiled surfaces to Masur—Veech volume of $\mathcal{H}(2g-2)$*, **GAFA** (2023).
2. V.Delecroix, M.Ebbens, F.Lazarus, I.Yakovlev, *Algorithms for Length Spectra of Combinatorial Tori*, **SoCG 2023**, **JoCG** (2024).
3. I.Yakovlev, *Metric ribbon graphs*, **thèse de doctorat** (2024).
4. E. Duryev, É. Goujard, I. Yakovlev, *Volumes of odd strata of quadratic differentials*, **soumis** (2025).

Surfaces de translation





Recollement de polygones le long de cotés égaux et parallèles.

$$\Updownarrow$$

Surface plate aux singularités coniques d'angles $2\pi k$ (et l'holonomie triviale).

Espaces de modules (paramètres)

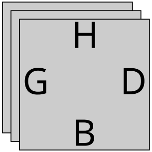
Strates des surfaces de translation :

$$\mathcal{H}(\underline{k}) = \mathcal{H}(k_1, \ldots, k_n) = \left\{ \begin{array}{c} \text{surfaces de translation aux} \\ \text{angles coniques } 2\pi(k_i + 1) \end{array} \right\}$$

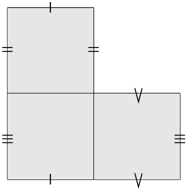
- variétés (localement $\mathbb{R}^n$);
- coordonnées : vecteurs des cotés des polygones;
- admettent une mesure finie naturelle (mesure de Masur–Veech).

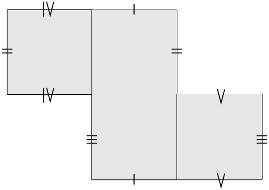
Surfaces à petits carreaux (SPC)

Calcul des volumes de Masur–Veech $\iff$ énumération asymptotique des "points entiers" dans les strates.



Règle de recollement :
$H \leftrightarrow B$, $G \leftrightarrow D$





Approches aux volumes de Masur–Veech

Théorie des représentations des groupes symétriques (Eskin-Okounkov '01), théorie de l'intersection (Sauvaget '18, Chen-Möller-Sauvaget-Zagier '20), approche combinatoire (Athreya-Eskin-Zorich '14, Delecroix-Goujard--Zograf-Zorich '21).

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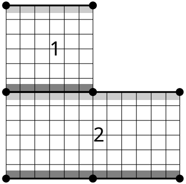
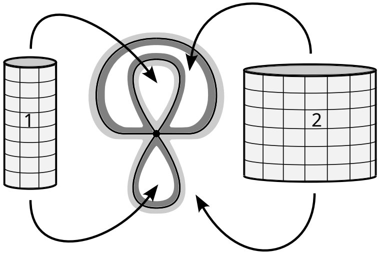
$$\text{SPC} = \text{cylindres} + \text{graphes en rubans métriques}$$





Comptage de graphes en rubans métriques

$\rightsquigarrow$ compter les graphes en rubans métriques de genre g avec n bords de longueurs $L_1, \ldots, L_n$.

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Théorème (Kontsevich '92)

Graphes trivalents $\Rightarrow$ polynôme en L_i dont les coefficients sont les nombres d'intersection $\int_{\overline{\mathcal{M}}_{g,n}} \psi_1^{d_1} \cdots \psi_n^{d_n}$.

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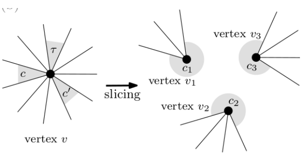
- **(Duryev-Goujard-Y. '25)** Graphes aux degrés de sommets impaires fixés.

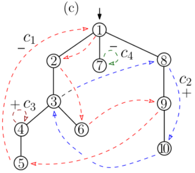
Outils : combinatoire bijective

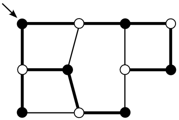
(Y. ’23) : bijection de Chapuy-Féray-Fusy ’13 entre les graphes en rubans à un bord et les arbres planes décorés.

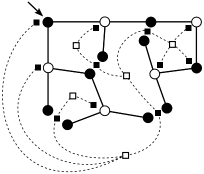
(Y. ’24) : bijection de Bernardi ’07 entre les graphes planaires munis d’un arbre couvrant et les paires d’arbres planaires.

(Duryev-Goujard-Y. ’25) : bijection à la Prüfer pour les arbres étiquetés.









Applications

Théorème (Y. '23)

Formule explicite pour la série génératrice bivariée $\mathcal{C}(t,u)$ des contributions $a_{g,n}$ des surfaces à petits carreaux avec n cylindres au volume de $\mathcal{H}(2g-2)$.

$\rightsquigarrow$ distribution des cylindres des SPC aléatoires dans $\mathcal{H}(2g-2)$ (travail en cours)

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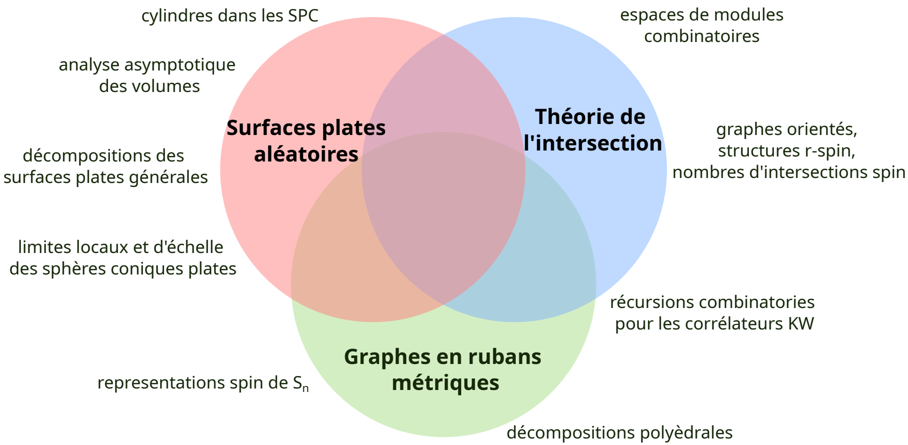
Théorème (Duryev-Goujard-Y. '25)

Récursion explicite pour les volumes des strates de différentielles quadratiques aux singularités impaires, qui fait intervenir les nombres d'intersection des classes psi et les cycles combinatoires de Kontsevich–Witten.

$\rightsquigarrow$ asymptotique des volumes pour $g \to \infty$ (travail en cours)

Projet de recherche

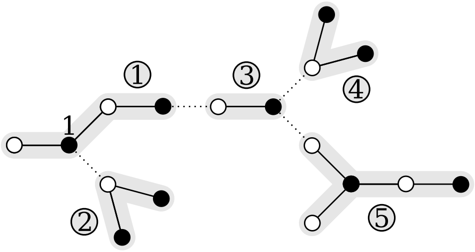


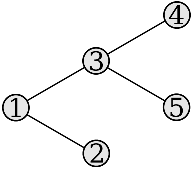
Intégration

- Thierry Monteil (surfaces de translation);
- Meltem Ünel (limites locaux/globaux d'arbres, des cartes combinatoires, géométrie hyperbolique);
- Lionel Pournin (géométrie polyèdrale, graphes des flips);
- Andrea Sportiello (systèmes intégrables, modèles de mécanique statistique);
- Axel Bacher, Cyril Banderier, Frédérique Bassino (combinatoire énumérative).

Slides cachés

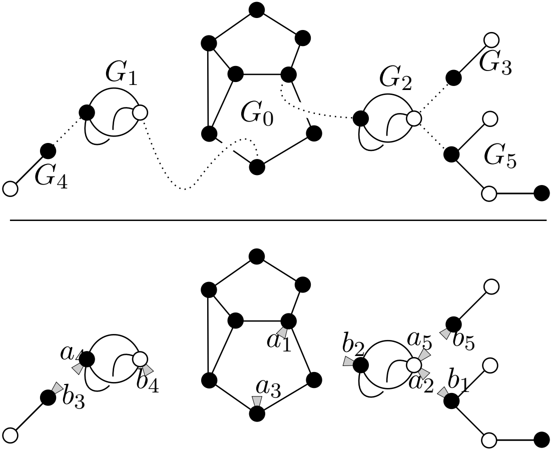
Computing degeneration coefficients on the walls





$$(k+l-2)! = p^{b_1,\dots,b_n}_{w_1,\dots,w_n} + \sum_{t=2}^{n} \frac{\color{red}{(k+l-2)_{t-2}}}{t!} \sum_{I_1,\dots I_t} \prod_{j=1}^{t} \left(\sum_{i \in I_j} (b_i + w_i) - 1 \right) p^{b_{I_j}}_{w_{I_j}},$$

Degenerations in the quadratic case



A joining is admissible if and only if for every $i = 1, \ldots, m$ the following holds:

- if all of the descendants of G_i have labels smaller then i, then the bridge joining G_i to its parent is in the black corner of G_i;
- otherwise, the bridge joining G_i to its parent and the bridge joining G_i to the subtree containing the descendant of G_i of maximal label are in the corners of G_i of different colors.

Cylinder contributions in $\mathcal{H}(2g-2)$

Theorem (Y., 2023)

The contribution of n-cylinder square-tiled surfaces to the volume of $\mathcal{H}(2g-2)$ is equal to $\frac{2(2\pi)^{2g}}{(2g-1)!}a_{g,n}$, where $a_{g,n} \in \mathbb{Q}$, and whose generating function $\mathcal{C}(t,u) = 1 + \sum_{g\geq 1}\left(\sum_{n=1}^{g} a_{g,n}u^n\right)(2g-1)t^{2g}$ satisfies for all $g \geq 0$

$$\frac{1}{(2g)!}[t^{2g}]\mathcal{C}(t,u)^{2g} = [t^{2g}]\left(\frac{t/2}{\sin(t/2)}\right)^u.$$

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- Faber-Pandharipande 2000: $\left(\frac{t/2}{\sin(t/2)}\right)^{u+1}$ is the generating function of Hodge integrals $\int_{\overline{\mathcal{M}}_{g,1}} \lambda_{g-i}\psi_1^{2g-2+i}$

Cylinder contributions in spin components of $\mathcal{H}(2g-2)$

Theorem (Y., 2024+)

The difference of the contributions of n-cylinder square-tiled surfaces to the volumes of even and odd spin subspaces of $\mathcal{H}(2g-2)$ is equal to $\frac{2(2\pi i)^{2g}}{(2g-1)!} d_{g,n}$, where $d_{g,n} \in \mathbb{Q}$, and whose generating function $\mathcal{D}(t,u) = 1 + \sum_{g \geq 1} \left(\sum_{n=1}^{g} d_{g,n} u^n \right) (2g-1) t^{2g}$ satisfies for all $k \geq 1$

$$\frac{1}{2k} [t^{2k}] \mathcal{D}(t,u)^{2k} = \frac{B_{2k}}{2^{k+1} k} u,$$

where B_{2k} is the $2k$-th Bernoulli number.

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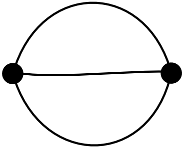
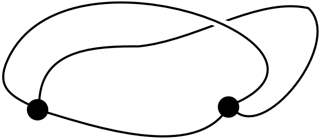
$u = 1$: formula from Chen, Möller, Sauvaget, Zagier.

Ribbon graphs (= combinatorial maps)

A ribbon graph is a graph (loops and multiple edges allowed) with a circular ordering of (half-)edges incident to every vertex.





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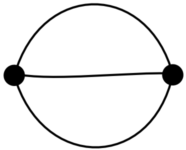

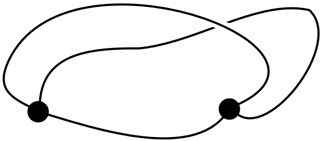

Equivalently, cellular embedding of a graph into a surface.

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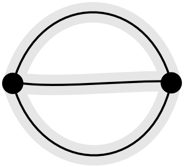

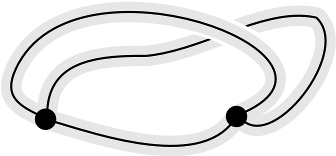

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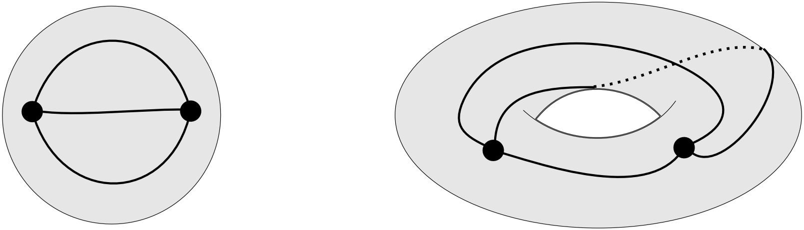

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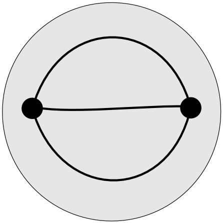

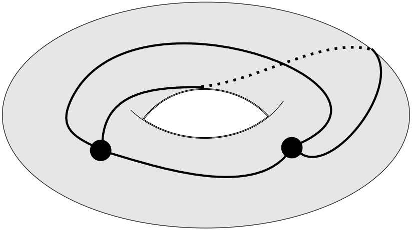

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Euler's formula: $|V(G)| - |E(G)| + |F(G)| = 2 - 2g.$

Metric ribbon graphs and counting functions

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For a ribbon graph G with n labeled faces and $L_1, \ldots, L_n \in \mathbb{Z}$, let

$$\mathcal{N}_G(L_1, \ldots, L_n) = \# \left\{ \begin{array}{c} \textit{integer} \text{ metrics on } G \text{ with perimeter of} \\ \text{the } i\text{-th face equal to } L_i \end{array} \right\}$$

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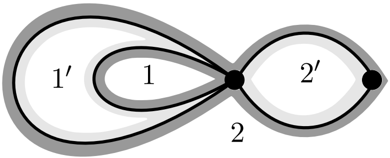
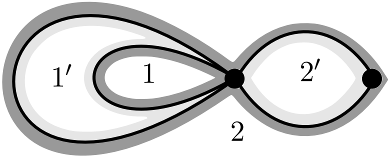
For a *face-bicolored* ribbon graph G with k black and l white faces, and $L_1, \ldots, L_k, L'_1, \ldots, L'_l \in \mathbb{Z}$, define analogously

$$\mathcal{N}_G(L_1, \ldots, L_k; L'_1, \ldots, L'_l).$$

Computing a counting function



Computing a counting function



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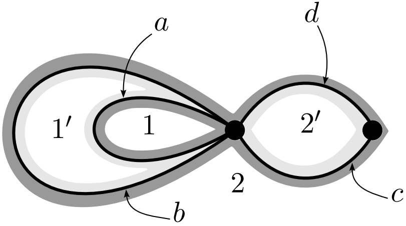

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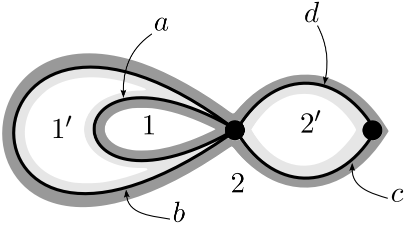

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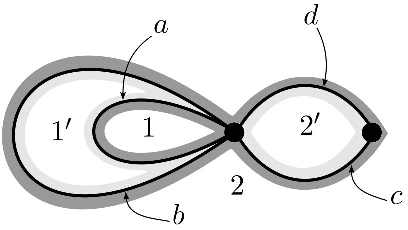

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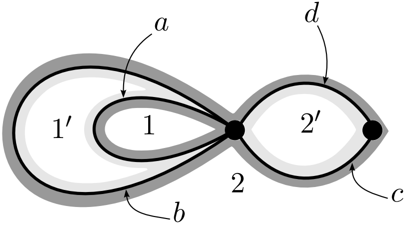

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Counting functions

Proposition

For any G the function $\mathcal{N}_G$ is piecewise (quasi-)polynomial. The regions of polynomiality are cut out by a certain hyperplane arrangement $\mathcal{H}_n$ $(\mathcal{H}_{k,l})$.

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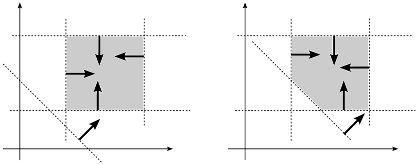

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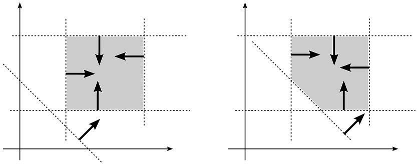

Note that the top-degree term of $\mathcal{N}_G$ gives the volume of this polytope!

Counting functions

Introduce the counting functions for families $\mathcal{RG}^d_{g,n}$ of ribbon graphs sharing the same genus g, number of faces n and vertex-degree profile d:

$$\mathcal{N}^d_{g,n}(L) = \sum_{G \in \mathcal{RG}^d_{g,n}} \frac{1}{|\operatorname{Aut}(G)|} \cdot \mathcal{N}_G(L).$$

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But sometimes the top-degree term is polynomial!

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Consider the families of *trivalent* ribbon graphs: $d = [3, \ldots, 3] = [3^{4g+2n-4}]$.

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Idea: polytopes corresponding to all graphs G can be glued together to form a space homeomorphic to $\overline{\mathcal{M}}_{g,n}$.

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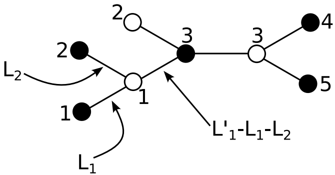
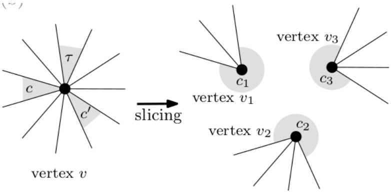
Idea: compute explicitly using a bijective approach!

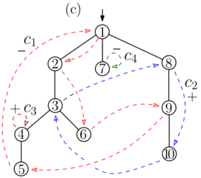
Example 2: one-vertex face-bicolored graphs

- $g = 0$: elementary count of metric plane trees;



- $g > 0$: reduce to case $g = 0$ using the bijection of Chapuy-Féray-Fusy '13 between 1-face maps and decorated plane trees, via vertex explosions.





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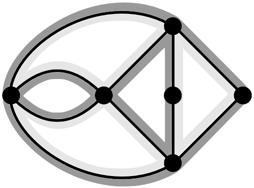

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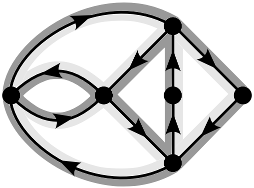

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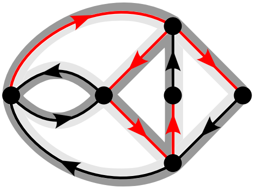

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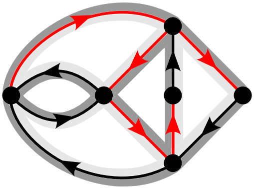

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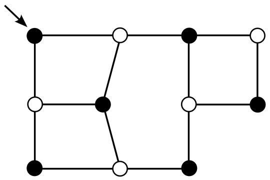

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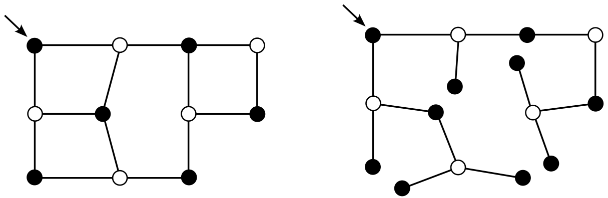

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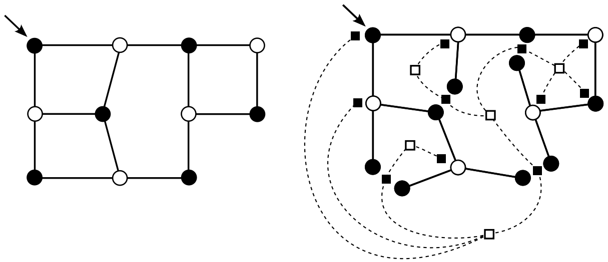

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- Formula $(k+l-2)! \cdot (L_1 + \ldots + L_k)^{\ell(d)-1}$ suggests a double counting argument: each tree contributes $(L_1 + \ldots + L_k)^{\ell(d)-1}$.
- It would be interesting to have such an argument for trivalent graphs $\Rightarrow$ combinatorics of the numbers $\int_{\overline{\mathcal{M}}_{g,n}} \psi_1^{\alpha_1} \cdots \psi_n^{\alpha_n}$.